



PREMIER UNIVERSITY CHATTOGRAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

FINAL PROJECT REPORT

COURSE NAME	Big Data Analytics Laboratory	
COURSE CODE	CSE 4346	
REPORT NAME	NYC Yellow Taxi Trip Data Analysis Using Hadoop MapReduce and Apache Pig.	
DATE OF REPORT	01-09-2026	
DATE OF SUBMISSION	15-09-2026	
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Introduction

Big Data technologies help process and analyze large datasets efficiently. This project analyzes the NYC Yellow Taxi Trip Dataset using Hadoop MapReduce and Apache Pig. The data is stored in HDFS and processed to obtain useful results. The dataset contains information about taxi trips, including trip distance, passenger count, fare, and payment type. Different data-processing techniques are applied to identify useful patterns and statistics. This project provides practical experience with distributed data processing using Hadoop.

Project Objectives

- Analyze NYC Yellow Taxi trip data using Hadoop.
- Perform 3 MapReduce operations using Java.
- Perform 5 Pig operations using Pig Latin.
- Store data and results in HDFS.
- Apply operations such as filtering, grouping, counting, sorting, and averaging.

Problem Statement

The NYC Yellow Taxi dataset contains information about taxi trips, including pickup and drop-off times, passenger count, trip distance, payment type, fare amount, tips, tolls, and total trip amount. Manually analyzing thousands or millions of records is inefficient. Therefore, Hadoop MapReduce and Apache Pig are used to process the dataset and extract useful information such as total trips, payment-type distribution, average distance, revenue, passenger distribution, and longest trips.

Dataset Description

The dataset used in this project is the **NYC Yellow Taxi Trip Data for March 2016**.

Dataset Link:

https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data?select=yellow_tripdata_2016-03.csv

The original dataset is approximately **1.8 GB** with **12.21 million records**. For this project, a reduced subset of approximately **600,000 records** was used to reduce processing time and resource requirements. The working dataset was analyzed using **Hadoop MapReduce and Apache Pig**.

Important Attributes

Attribute	Description
VendorID	Identifier of the taxi technology/vendor provider
pickup_datetime	Date and time when the trip started

dropoff_datetime	Date and time when the trip ended
passenger_count	Number of passengers
trip_distance	Distance travelled during the trip
pickup_longitude	Pickup longitude
pickup_latitude	Pickup latitude
RatecodeID	Rate code used for the trip
store_and_fwd_flag	Indicates whether the trip record was stored before forwarding
dropoff_longitude	Drop-off longitude
dropoff_latitude	Drop-off latitude
payment_type	Method used to pay for the trip
fare_amount	Base fare amount
extra	Additional charges
mta_tax	MTA tax
tip_amount	Tip amount
tolls_amount	Toll charges
improvement_surcharge	Improvement surcharge
total_amount	Total amount charged

Summary of Data Analysis

The NYC Yellow Taxi dataset was analyzed using Apache Hadoop MapReduce and Apache Pig in a Windows-based Hadoop environment. The dataset was uploaded to HDFS and processed using distributed data-processing techniques.

MapReduce Operations

1. **Total Taxi Trips** – Counted valid taxi trips (~12.21 million).
2. **Trips by Payment Type** – Counted trips for each payment method.
3. **Average Trip Distance** – Calculated the average distance of taxi trips.

Pig Operations

1. **Total Revenue** – Calculated overall taxi revenue.
2. **Average Fare by Payment Type** – Compared average fares across payment methods.
3. **Trips by Passenger Count** – Counted trips based on passenger numbers.
4. **Top 10 Longest Trips** – Identified the longest taxi journeys.
5. **Payment Type Analysis** – Calculated total trips, revenue, and average distance for each payment type.

System Environment

1. Operating System: Windows 10
2. Java: JDK 1.8
3. Hadoop: 3.2.4
4. Apache Pig: 0.18.0
5. Processing Mode: Hadoop MapReduce
6. Storage: Hadoop Distributed File System (HDFS)
7. Shell: Windows PowerShell

Prepare and Verify the Environment

Before processing the transaction dataset, the required Hadoop ecosystem components must be available. Java is required for the Hadoop platform, Hadoop provides HDFS and YARN services, and Apache Pig provides the data-flow processing layer used later in the project. The project work is organized on the D drive. The project folders contain the dataset, MapReduce source files, Pig scripts, and project outputs. Software installations remain in their configured installation directories.

Java version verification

```
C:\Windows\system32>java -version
java version "1.8.0_411"
Java(TM) SE Runtime Environment (build 1.8.0_411-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.411-b09, mixed mode)
```

Hadoop & Pig version verification

```
C:\Windows\system32>hadoop -version
java version "1.8.0_411"
Java(TM) SE Runtime Environment (build 1.8.0_411-b09)
Java HotSpot(TM) 64-Bit Server VM (build 25.411-b09, mixed mode)

C:\Windows\system32>pig -version
Apache Pig version 0.18.0 (r1926484)
compiled Jun 16 2025, 10:38:21
```

Start Hadoop Services

The Hadoop services must be running before HDFS and YARN-based processing can be performed. The running processes confirm that the single-node Hadoop environment is active and ready for distributed jobs. The main processes used in the project are NameNode for HDFS metadata management, DataNode for data block storage, ResourceManager for YARN resource management, and NodeManager for container execution.

Hadoop services running in jps

```
C:\Windows\system32>jps
13504 ResourceManager
14544 DataNode
5844 NameNode
10712 Jps
8520 NodeManager
```

YARN node in running state

```
C:\Windows\system32>yarn node -list
2026-09-14 16:40:48,756 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
Total Nodes:1
  Node-Id          Node-State Node-Http-Address      Number-of-Running-Containers
DESKTOP-JVF3IDO.ms Node-Id          Node-State Node-Http-Address      RUNNING DESKTOP-JVF3IDO.ms
DESKTOP-JVF3IDO.ms home.net:53226 home.net:8042

C:\Windows\system32>tasklist | findstr /I "java"
java.exe           5844 Console           5      224,156 K
java.exe           14544 Console          5      190,260 K
java.exe           13504 Console          5      230,156 K
java.exe           8520 Console           5      166,988 K
```

Hadoop Web Interface

Hadoop's web interface can be used to browse the HDFS directory structure and inspect output folders. The same interface is useful later for demonstrating the results of the MapReduce and Pig tasks.

Hadoop localhost HDFS browser showing the transaction directory.

Browse Directory

/

Go!

Show

25

entries

Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Aug 03 22:22	0	0 B	BDA6	
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 13 14:23	0	0 B	NYC-Yellow-Taxi-Analysis	
☐	drwx-----	mdsakib	supergroup	0 B	Sep 14 01:00	0	0 B	tmp	
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Aug 30 16:26	0	0 B	user	
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Aug 25 00:09	0	0 B	youtube	

Showing 1 to 5 of 5 entries

Previous

1

Next

Map Reduce Analysis :

Merge Files : Merges all the CSV Files into one and stores it into HDFS

Script :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -mkdir -p /NYC-Yellow-Taxi-Analysis/input
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis
Found 1 items
drwxr-xr-x - mdsakib supergroup 0 2026-09-13 01:10 /NYC-Yellow-Taxi-Analysis/input
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis
Found 1 items
drwxr-xr-x - mdsakib supergroup 0 2026-09-13 01:10 /NYC-Yellow-Taxi-Analysis/input
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -put ".\dataset\yellow_tripdata_2016_03_6000.csv" /NYC-Yellow-Taxi-Analysis/input/
put: '.\dataset\yellow_tripdata_2016_03_6000.csv': No such file or directory
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -put ".\dataset\yellow_tripdata_2016_03_6000.csv" /NYC-Yellow-Taxi-Analysis/input/
put: '.\dataset\yellow_tripdata_2016_03_6000.csv': No such file or directory
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -ls -h /NYC-Yellow-Taxi-Analysis/input
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 01:39 /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016-03-noheader.csv._COPYING_
-rw-r--r-- 1 mdsakib supergroup 1.8 G 2026-09-13 00:57 /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016-03.csv
NYC-Yellow-Taxi-Analysis> ^C
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> Test-Path ".\dataset\yellow_tripdata_2016_03_6000.csv"
True
hdfs dfs -rm -r /NYC-Yellow-Taxi-Analysis/input/*PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis>
Deleted /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016-03-noheader.csv._COPYING_
Deleted /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016-03.csv
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -put ".\dataset\yellow_tripdata_2016_03_6000.csv" /NYC-Yellow-Taxi-Analysis/input/
put: '.\dataset\yellow_tripdata_2016_03_6000.csv': No such file or directory
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -put "${PWD}\dataset\yellow_tripdata_2016_03_6000.csv" /NYC-Yellow-Taxi-Analysis/input/
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -ls -h /NYC-Yellow-Taxi-Analysis/input
Found 1 items
-rw-r--r-- 1 mdsakib supergroup 919.5 K 2026-09-13 13:54 /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016_03_6000.csv
```

Execution :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016_03_6000.csv | Select-Object -First 5
VendorID,trip_pickup_datetime,trip_dropoff_datetime,passenger_count,trip_distance,pickup_longitude,pickup_latitude,RatecodeID,store_and_fwd_flag,dropoff_longitude,dropoff_latitude,payment_type
fare_amount,extra,mta_tax,tip_amount,tolls_amount,improvement_surchage,total_amount
1,2016-03-01 00:00:00,2016-03-01 00:07:55,1,2.50,-73.97674560546875,40.76515197539862,1,N,-74.004264831542969,40.746128882275391,1,9,0.5,0.5,2.05,0.0,3,12.35
1,2016-03-01 00:00:00,2016-03-01 00:11:06,1,2.90,-73.983482360839844,40.767925262451172,1,N,-74.005943298339844,40.733165740966797,1,11,0.5,0.5,3.05,0.0,3,15.35
2,2016-03-01 00:00:00,2016-03-01 00:31:06,2,19.98,-73.782020568847656,40.644809722900391,1,N,-73.974540710449219,40.675769805908203,1,54,5,0.5,0.5,8,0.3,63.8
2,2016-03-01 00:00:00,2016-03-01 00:00:00,3,10.78,-73.863418579101562,40.769813537597656,1,N,-73.969650268554688,40.757766723632812,1,31,5,0.0,5,3.78,5.54,0.3,41.62
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016_03_6000.csv | Select-Object -Skip 1 | hdfs dfs -put -^C
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> hdfs dfs -ls -h /NYC-Yellow-Taxi-Analysis/input
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 919.2 K 2026-09-13 13:56 /NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv
-rw-r--r-- 1 mdsakib supergroup 919.5 K 2026-09-13 13:54 /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016_03_6000.csv
cd "C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis"
```

Part A · Implement MapReduce (3 jobs)

MR-1 : Total number of taxi trips

TotalTripsMapper.java :

```
*TotalTripsMapper.java - Notepad
File Edit Format View Help
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;

public class TotalTripsMapper
    extends Mapper<LongWritable, Text, Text, IntWritable> {

    private static final IntWritable ONE =
        new IntWritable(1);

    private static final Text TOTAL_TRIPS =
        new Text("Total Trips");

    @Override
    public void map(
        LongWritable key,
        Text value,
        Context context)
        throws IOException, InterruptedException {

        String line = value.toString().trim();

        // Ignore CSV header
        if (line.startsWith("VendorID")) {
            return;
        }

        // Ignore empty lines
        if (line.isEmpty()) {
            return;
        }

        context.write(TOTAL_TRIPS, ONE);
    }
}
```

TotalTripsReducer.java :

```
*TotalTripsReducer.java - Notepad
File Edit Format View Help
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class TotalTripsReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(
        Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int total = 0;

        for (IntWritable value : values) {
            total += value.get();
        }

        context.write(
            key,
            new IntWritable(total)
        );
    }
}
```

TotalTripsDriver.java :



*TotalTripsDriver.java - Notepad

```
File Edit Format View Help
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;

import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class TotalTripsDriver {

    public static void main(String[] args)
        throws Exception {

        if (args.length != 2) {

            System.err.println(
                "Usage: TotalTripsDriver <input> <output>"
            );

            System.exit(2);

        }

        Configuration conf =
            new Configuration();

        Job job = Job.getInstance(
            conf,
            "NYC Taxi - Total Trips"
        );

        job.setJarByClass(
            TotalTripsDriver.class
        );

        job.setMapperClass(
            TotalTripsMapper.class
        );

        job.setReducerClass(
            TotalTripsReducer.class
        );

        job.setOutputKeyClass(
            Text.class
        );

        job.setOutputValueClass(
            IntWritable.class
        );

        FileInputFormat.addInputPath(
            job,
            new Path(args[0])
        );

        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );

        System.exit(
            job.waitForCompletion(true)
                ? 0
                : 1
        );

    }
}
```

Execution :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR1_TotalTrips> hadoop jar total-trips.jar
C-Yellow-Taxi-Analysis/output/mr1
2026-09-13 14:23:39,205 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 14:23:40,339 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not perform
y this.
2026-09-13 14:23:40,370 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop
2026-09-13 14:23:40,669 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 14:23:40,786 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-13 14:23:41,039 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789285363593_0001
2026-09-13 14:23:41,039 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 14:23:41,320 INFO conf.Configuration: resource-types.xml not found
2026-09-13 14:23:41,320 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 14:23:41,928 INFO impl.YarnClientImpl: Submitted application application_1789285363593_0001
2026-09-13 14:23:42,007 INFO mapreduce.Job: The url to track the job: http://DESKTOP-JVF3IDO:8088/proxy/ap
2026-09-13 14:23:42,007 INFO mapreduce.Job: Running job: job_1789285363593_0001
2026-09-13 14:23:54,443 INFO mapreduce.Job: Job job_1789285363593_0001 running in uber mode : false
2026-09-13 14:23:54,443 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 14:24:02,771 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 14:24:11,898 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 14:24:11,899 INFO mapreduce.Job: Job job_1789285363593_0001 completed successfully
2026-09-13 14:24:12,015 INFO mapreduce.Job: Counters: 54
```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR1_TotalTrips> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/mr1/part-r-00000
Total Trips      6000
```

Block information

Block 0

Block ID: 1073742072

Block Pool ID: BP-1863203093-192.168.0.101-1785770156191

Generation Stamp: 1248

Size: 17

Availability:

- DESKTOP-JVF3IDO.mshome.net

File contents

Total Trips	6000
-------------	------

MR-2 : Trips by Payment Type

PaymentTypeMapper.java

```
*PaymentTypeMapper.java - Notepad
File Edit Format View Help
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class PaymentTypeMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private static final IntWritable ONE =
        new IntWritable(1);

    @Override
    public void map(
        Object key,
        Text value,
        Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        if (line.startsWith("VendorID")) {
            return;
        }
        String[] fields =
            line.split(",", -1);

        if (fields.length < 12) {
            return;
        }

        String paymentType =
            fields[11].trim();

        if (!paymentType.isEmpty()) {
            context.write(
                new Text(paymentType),
                ONE
            );
        }
    }
}
```

PaymentTypeReducer.java

```
*PaymentTypeReducer.java - Notepad
File Edit Format View Help
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;

public class PaymentTypeReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    @Override
    public void reduce(
        Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int total = 0;

        for (IntWritable value : values) {
            total += value.get();
        }

        context.write(
            key,
            new IntWritable(total)
        );
    }
}
```

PaymentTypeDriver.java

```
*PaymentTypeDriver.java - Notepad
File Edit Format View Help
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class PaymentTypeDriver {
    public static void main(String[] args)
        throws Exception {
        if (args.length != 2) {
            System.err.println(
                "Usage: PaymentTypeDriver <input> <output>"
            );System.exit(2);
        }
        Configuration conf =
            new Configuration();

        Job job = Job.getInstance(
            conf,
            "NYC Taxi - Payment Type Analysis"
        );
        job.setJarByClass(
            PaymentTypeDriver.class
        );
        job.setMapperClass(
            PaymentTypeMapper.class
        );
        job.setReducerClass(
            PaymentTypeReducer.class
        );
        job.setOutputKeyClass(
            Text.class
        );
        job.setOutputValueClass(
            IntWritable.class
        );
        FileInputFormat.addInputPath(
            job,
            new Path(args[0])
        );
        FileOutputFormat.setOutputPath(
            job,
            new Path(args[1])
        );
        System.exit(
            job.waitForCompletion(true)
                ? 0
                : 1
        );
    }
}
```

Execution :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR2_PaymentType> hadoop jar payment-type.jar PaymentTypeDriver /NYC-Yellow-Taxi-Analysis/input/yellow_tripdata_2016_03_0800.csv
/NYC-Yellow-Taxi-Analysis/output/mr2
2026-09-13 16:10:13,665 INFO client.RMProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 16:10:14,833 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-09-13 16:10:14,862 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/mdsakib/.staging/job_1789285363593_0002
2026-09-13 16:10:15,282 INFO InputFileInputFormat: total input files to process : 1
2026-09-13 16:10:15,388 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-13 16:10:15,590 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789285363593_0002
2026-09-13 16:10:15,594 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 16:10:15,862 INFO conf.Configuration: resource-types.xml not found
2026-09-13 16:10:15,863 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 16:10:15,999 INFO impl.YarnClientImpl: Submitted application application_1789285363593_0002
2026-09-13 16:10:16,077 INFO mapreduce.Job: The url to track the job: http://DESKTOP-JVF3ID0:8088/proxy/application_1789285363593_0002/
2026-09-13 16:10:16,079 INFO mapreduce.Job: Running job: job_1789285363593_0002
2026-09-13 16:10:20,542 INFO mapreduce.Job: Job Job_1789285363593_0002 running in uber mode : false
2026-09-13 16:10:28,545 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 16:10:35,744 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 16:10:42,840 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 16:10:42,854 INFO mapreduce.Job: Job Job_1789285363593_0002 completed successfully
2026-09-13 16:10:43,020 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=48006
  FILE: Number of bytes written=574115
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=941711
  HDFS: Number of bytes written=18
  HDFS: Number of read operations=8
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0
```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR2_PaymentType> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/mr2/part-r-00000
1      4387
2      1607
3        6
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR2_PaymentType>
```

File contents

```
1      4387
2      1607
3        6
```

MR-3 : Calculate average taxi trip distance

AverageDistanceMapper.java

```
*AverageDistanceMapper.java - Notepad
File Edit Format View Help
import java.io.IOException;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Mapper;
public class AverageDistanceMapper
    extends Mapper<LongWritable, Text, Text, Text> {
    private static final Text TOTAL =
        new Text("TOTAL");
    @Override
    public void map(
        LongWritable key,
        Text value,
        Context context)
        throws IOException, InterruptedException {
        String line = value.toString();
        if (line.startsWith("VendorID")) {
            return;
        }
        String[] fields =
            line.split(",", -1);
        if (fields.length < 5) {
            return;
        }
        try {
            double distance =
                Double.parseDouble(
                    fields[4].trim()
                );
            if (distance >= 0) {
                context.write(
                    TOTAL,
                    new Text(
                        distance + ",1"
                    )
                );
            }
        } catch (NumberFormatException e) {
            // Ignore invalid values
        }
    }
}
```

AverageDistanceReducer.java

```
*AverageDistanceReducer.java - Notepad
File Edit Format View Help
import java.io.IOException;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Reducer;
public class AverageDistanceReducer
    extends Reducer<Text, Text, Text, Text> {
    @Override
    public void reduce(
        Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {
        double totalDistance = 0.0;
        long totalTrips = 0;
        for (Text value : values) {
            String[] parts =
                value.toString().split(",");

            if (parts.length != 2) {
                continue;
            }
            try {

                double distance =Double.parseDouble(parts[0]);
                long count =Long.parseLong(parts[1]);
                totalDistance += distance;
                totalTrips += count;
            } catch (NumberFormatException e) {
                // Ignore invalid values
            }
        }

        if (totalTrips > 0) {
            double average =
                totalDistance / totalTrips;
            context.write(
                new Text(
                    "Average Trip Distance"
                ),
                new Text(String.format("%.4f miles",average))
            );
        }
    }
}
```

AverageDistanceDriver.java

```
*AverageDistanceDriver.java - Notepad
File Edit Format View Help
import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;
public class AverageDistanceDriver {
    public static void main(String[] args)
        throws Exception {
        if (args.length != 2) {
            System.err.println(
                "Usage: AverageDistanceDriver <input> <output>"
            );
            System.exit(2);
        }
        Configuration conf =new Configuration();
        Job job = Job.getInstance(conf,"NYC Taxi - Average Distance");
        job.setJarByClass(AverageDistanceDriver.class);
        job.setMapperClass(AverageDistanceMapper.class);
        job.setReducerClass(AverageDistanceReducer.class);
        job.setOutputKeyClass(Text.class);
        job.setOutputValueClass(Text.class);
        FileInputFormat.addInputPath(job,new Path(args[0]));
        FileOutputFormat.setOutputPath(job,new Path(args[1]));
        System.exit(job.waitForCompletion(true)
            ? 0
            : 1
        );
    }
}
```

Execution :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR3_AverageDistance> hadoop jar average03_6000.csv /NYC-Yellow-Taxi-Analysis/output/mr3
2026-09-13 16:34:06,217 INFO client.RMPProxy: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 16:34:07,077 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed by this.
2026-09-13 16:34:07,097 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/had
2026-09-13 16:34:07,432 INFO input.FileInputFormat: Total input files to process : 1
2026-09-13 16:34:07,530 INFO mapreduce.JobSubmitter: number of splits:1
2026-09-13 16:34:07,709 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789285363593_0003
2026-09-13 16:34:07,714 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-13 16:34:07,960 INFO conf.Configuration: resource-types.xml not found
2026-09-13 16:34:07,961 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-13 16:34:08,072 INFO impl.YarnClientImpl: Submitted application application_1789285363593_0003
2026-09-13 16:34:08,143 INFO mapreduce.Job: The url to track the job: http://DESKTOP-JVF3IDO:8088/proxy
2026-09-13 16:34:08,144 INFO mapreduce.Job: Running job: job_1789285363593_0003
2026-09-13 16:34:18,436 INFO mapreduce.Job: Job job_1789285363593_0003 running in uber mode : false
2026-09-13 16:34:18,441 INFO mapreduce.Job: map 0% reduce 0%
2026-09-13 16:34:26,695 INFO mapreduce.Job: map 100% reduce 0%
2026-09-13 16:34:34,845 INFO mapreduce.Job: map 100% reduce 100%
2026-09-13 16:34:34,859 INFO mapreduce.Job: Job job_1789285363593_0003 completed successfully
2026-09-13 16:34:34,986 INFO mapreduce.Job: Counters: 54
```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR3_AverageDistance> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/mr3/part-r-00000
Average Trip Distance 2.9481 miles
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\mapreduce\MR3_AverageDistance>
```

File contents

Average Trip Distance	2.9481 miles
-----------------------	--------------

Part B · Apache Pig Analysis (5 jobs)

Pig-1 : Calculate the total revenue generated from all valid taxi trips.

p1_total_revenue Script :

```
p1_total_revenue - Notepad
File Edit Format View Help
|taxi_data = LOAD '/NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv'
|USING PigStorage(',')
|AS (
|    VendorID:int,
|    pickup_datetime:chararray,
|    dropoff_datetime:chararray,
|    passenger_count:int,
|    trip_distance:double,
|    pickup_longitude:double,
|    pickup_latitude:double,
|    RatecodeID:int,
|    store_and_fwd_flag:chararray,
|    dropoff_longitude:double,
|    dropoff_latitude:double,
|    payment_type:int,
|    fare_amount:double,
|    extra:double,
|    mta_tax:double,
|    tip_amount:double,
|    tolls_amount:double,
|    improvement_surcharge:double,
|    total_amount:double
|);
|
|valid_data = FILTER taxi_data BY total_amount >= 0;
|
|grouped_data = GROUP valid_data ALL;
|
|total_revenue = FOREACH grouped_data GENERATE
|    SUM(valid_data.total_amount) AS total_revenue;
|
|DUMP total_revenue;
|
|STORE total_revenue
|INTO '/NYC-Yellow-Taxi-Analysis/output/pig/P1_TotalRevenue'
|USING PigStorage(',');
|
```

Execution :

```

2026-09-13 17:08:18,704 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Submitting tokens for job: job_1789285363593_0004
2026-09-13 17:08:18,711 [JobControl] INFO org.apache.hadoop.mapreduce.JobSubmitter - Executing with tokens: []
2026-09-13 17:08:18,919 [JobControl] INFO org.apache.hadoop.mapred.YARNRunner - Job jar is not present. Not adding any jar to the list of resources.
2026-09-13 17:08:18,992 [JobControl] INFO org.apache.hadoop.conf.Configuration - resource-types.xml not found
2026-09-13 17:08:18,993 [JobControl] INFO org.apache.hadoop.yarn.util.resource.ResourceUtils - Unable to find 'resource-types.xml'.
2026-09-13 17:08:18,969 [JobControl] INFO org.apache.hadoop.yarn.client.api.impl.YarnClientImpl - Submitted application application_1789285363593_0004
2026-09-13 17:08:19,123 [JobControl] INFO org.apache.hadoop.mapreduce.Job - The url to track the job: http://DESKTOP-1VF3D00:8088/proxy/application_1789285363593_0004/
2026-09-13 17:08:19,123 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.HadoopJobId - HadoopJobId: job_1789285363593_0004
2026-09-13 17:08:19,124 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Processing aliases grouped_data,taxi_data,total_revenue,valid_data
2026-09-13 17:08:19,127 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: taxi_data[1,12],taxi_data[-1,-1],valid_data[25,13],
total_revenue[29,16],grouped_data[27,15]
2026-09-13 17:08:19,146 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-13 17:08:19,146 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789285363593_0004]
2026-09-13 17:08:41,570 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-13 17:08:41,570 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789285363593_0004]
2026-09-13 17:08:41,570 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789285363593_0004]
2026-09-13 17:08:54,321 [main] INFO org.apache.hadoop.yarn.client.HMPProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 17:08:54,348 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to Job history server
2026-09-13 17:08:57,391 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 17:25:40,065 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 17:25:43,125 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 17:25:46,181 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 17:25:49,240 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 17:25:51,377 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-13 17:25:51,377 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 17:25:51,426 [main] INFO org.apache.pig.Main - Pig script completed in 17 minutes, 37 seconds and 653 milliseconds (1067553 ms)
PS C:\Users\Wd Sakib\Desktop\WVC-Yellow-Taxi-Analysis> pig\PigTotalRevenue> hdfs dfs -ls /NVC-Yellow-Taxi-Analysis/output/pig/PigTotalRevenue
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 17:17 /NVC-Yellow-Taxi-Analysis/output/pig/PigTotalRevenue/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup 18 2026-09-13 17:17 /NVC-Yellow-Taxi-Analysis/output/pig/PigTotalRevenue/part-r-000000

```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig\PI_TotalRevenue> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/PI_TotalRevenue
Found 2 items
-rw-r--r-- 1 mdsakib supergroup          0 2026-09-13 17:17 /NYC-Yellow-Taxi-Analysis/output/pig/PI_TotalRevenue/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup    18 2026-09-13 17:17 /NYC-Yellow-Taxi-Analysis/output/pig/PI_TotalRevenue/part-r-00000
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig\PI_TotalRevenue> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/pig/PI_TotalRevenue/part-r-00000
91415.00000000514
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig\PI_TotalRevenue>
```

File contents

91415.00000000514

Pig-2 : Find average fare by payment type

p2_average_fare Script :

```
*p2_average_fare - Notepad
File Edit Format View Help
taxi_data = LOAD '/NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv'
USING PigStorage(',')
AS (
  VendorID:int,
  pickup_datetime:chararray,
  dropoff_datetime:chararray,
  passenger_count:int,
  trip_distance:double,
  pickup_longitude:double,
  pickup_latitude:double,
  RatecodeID:int,
  store_and_fwd_flag:chararray,
  dropoff_longitude:double,
  dropoff_latitude:double,
  payment_type:int,
  fare_amount:double,
  extra:double,
  mta_tax:double,
  tip_amount:double,
  tolls_amount:double,
  improvement_surcharge:double,
  total_amount:double
);

valid_data = FILTER taxi_data BY fare_amount >= 0;
grouped_payment = GROUP valid_data BY payment_type;
average_fare = FOREACH grouped_payment GENERATE
  group AS payment_type,
  AVG(valid_data.fare_amount) AS average_fare;

DUMP average_fare;

STORE average_fare
INTO '/NYC-Yellow-Taxi-Analysis/output/pig/P2_AverageFare'
USING PigStorage(',')];
```

Execution :

```
average_fare[29,15],grouped_payment[27,18] C: average_fare[29,15],grouped_payment[27,18] R: average_fare[29,15]
2026-09-13 20:34:08,095 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-13 20:34:08,095 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0001]
2026-09-13 20:34:30,608 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-13 20:34:30,608 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0001]
2026-09-13 20:34:40,382 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0001]
2026-09-13 20:34:43,390 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8082
2026-09-13 20:34:43,406 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 20:34:46,444 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:34:49,488 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:34:52,530 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:34:55,574 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:34:58,616 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:01,680 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:04,727 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:07,782 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:10,842 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:13,896 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:16,975 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 20:35:19,116 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:21,620 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:24,675 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 6 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:27,719 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:30,764 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:33,829 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 20:35:35,970 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-13 20:35:35,970 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 20:35:36,018 [main] INFO org.apache.pig.Main - Pig script completed in 17 minutes, 35 seconds and 395 milliseconds (1055386 ms)
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig(P2_AverageFare) hdfs dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/P2_AverageFare
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 20:43 /NYC-Yellow-Taxi-Analysis/output/pig/P2_AverageFare/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup 50 2026-09-13 20:43 /NYC-Yellow-Taxi-Analysis/output/pig/P2_AverageFare/part-r-00000
```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P2_AverageFare> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/pig/P2_AverageFare/part-r-00000
1,12.865910645087759
2,10.571773490976975
3,13.75
```

File contents	
1,12.865910645087759	
2,10.571773490976975	
3,13.75	

Pig-3 : Find number of trips by passenger count

p3_passenger_count Script :

 p3_passenger_count - Notepad

File Edit Format View Help

```
taxi_data = LOAD '/NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv'
USING PigStorage(',')
AS (
  VendorID:int,
  pickup_datetime:chararray,
  dropoff_datetime:chararray,
  passenger_count:int,
  trip_distance:double,
  pickup_longitude:double,
  pickup_latitude:double,
  RatecodeID:int,
  store_and_fwd_flag:chararray,
  dropoff_longitude:double,
  dropoff_latitude:double,
  payment_type:int,
  fare_amount:double,
  extra:double,
  mta_tax:double,
  tip_amount:double,
  tolls_amount:double,
  improvement_surcharge:double,
  total_amount:double
);

valid_data = FILTER taxi_data
BY passenger_count > 0;

grouped_passengers =
  GROUP valid_data BY passenger_count;

passenger_trips =
  FOREACH grouped_passengers GENERATE
    group AS passenger_count,
    COUNT(valid_data) AS total_trips;

ordered_result =
  ORDER passenger_trips BY passenger_count ASC;

DUMP ordered_result;

STORE ordered_result
INTO '/NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount'
USING PigStorage(',')|
```

Execution :

```
2026-09-13 21:42:15,676 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: taxi_data[1,12],taxi_data[-1,-1],valid_data[25,13],
passenger_trips[32,4],grouped_passengers[29,4] C: passenger_trips[32,4],grouped_passengers[29,4] R: passenger_trips[32,4]
2026-09-13 21:42:15,690 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 9% complete
2026-09-13 21:42:15,690 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_17893099080849_0006]
2026-09-13 21:42:40,248 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 16% complete
2026-09-13 21:42:40,252 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_17893099080849_0006]
2026-09-13 21:42:49,961 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 33% complete
2026-09-13 21:42:49,964 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_17893099080849_0006]
2026-09-13 21:42:56,093 [main] INFO org.apache.hadoop.yarn.client.RMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 21:42:56,107 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 21:42:59,169 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 21:43:02,207 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 21:43:05,275 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 21:43:08,313 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 21:43:11,355 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 21:43:14,387 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 5 time(s); retry policy is RetryUpToMaximumCountWithFixedSI
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
```

```

eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 22:08:56,777 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 22:08:59,828 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 22:09:02,893 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-13 22:09:05,050 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-13 22:09:05,050 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 22:09:05,072 [main] INFO org.apache.pig.Main - Pig script completed in 53 minutes, 28 seconds and 329 milliseconds (3208329 ms)
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P3_PassengerCount> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 21:54 /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup 37 2026-09-13 21:54 /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount/part-r-000000

```

Output:

```

PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P3_PassengerCount> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 21:54 /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup 37 2026-09-13 21:54 /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount/part-r-000000
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P3_PassengerCount> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/pig/P3_PassengerCount/part-r-000000
1,3837
2,839
3,240
4,110
5,572
6,402

```

File contents

```

1,3837
2,839
3,240
4,110
5,572
6,402

```

Pig-4 : Find top 10 longest trips

p4_longest_trips Script :

```

p4_longest_trips - Notepad
File Edit Format View Help
taxi_data = LOAD '/NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv'
USING PigStorage(',')
AS (
    VendorID:int,
    pickup_datetime:chararray,
    dropoff_datetime:chararray,
    passenger_count:int,
    trip_distance:double,
    pickup_longitude:double,
    pickup_latitude:double,
    RatecodeID:int,
    store_and_fwd_flag:chararray,
    dropoff_longitude:double,
    dropoff_latitude:double,
    payment_type:int,
    fare_amount:double,
    extra:double,
    mta_tax:double,
    tip_amount:double,
    tolls_amount:double,
    improvement_surcharge:double,
    total_amount:double
);

valid_trips = FILTER taxi_data
BY trip_distance > 0 AND trip_distance < 100;

sorted_trips =
    ORDER valid_trips BY trip_distance DESC;

top_10 =
    LIMIT sorted_trips 10;

result = FOREACH top_10 GENERATE
    pickup_datetime,
    dropoff_datetime,
    passenger_count,
    trip_distance,
    fare_amount,
    total_amount;

DUMP result;

STORE result
INTO '/NYC-Yellow-Taxi-Analysis/output/pig/P4_LongestTrips'
USING PigStorage(',')];

```

Execution :

```
2026-09-13 23:02:22,683 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - detailed locations: M: taxi_data[1,12],taxi_data[-1,-1],valid_trips[25,14]
C: R:
2026-09-13 23:02:22,691 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-13 23:02:22,692 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309908049_0013]
2026-09-13 23:02:45,113 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 12% complete
2026-09-13 23:02:45,113 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309908049_0013]
2026-09-13 23:02:48,146 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 25% complete
2026-09-13 23:02:48,159 [main] INFO org.apache.hadoop.yarn.client.NMProxy - Connecting to ResourceManager at /0.0.0.0:8032
2026-09-13 23:02:48,166 [main] INFO org.apache.hadoop.mapred.ClientServiceDelegate - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-13 23:02:51,206 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:02:54,270 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:02:57,327 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 2 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:00,390 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 3 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:03,434 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 4 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:06,562 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 50% complete
2026-09-13 23:03:18,539 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 7 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:21,562 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:24,604 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSl
eep(maxRetries=10, sleepTime=1000 MILLISECOND)
2026-09-13 23:03:25,776 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-13 23:03:25,777 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Success!
2026-09-13 23:03:26,817 [main] INFO org.apache.pig.Main - Pig script completed in 1 hour, 9 minutes, 51 seconds and 476 milliseconds (4191476 ms)
```

Output :

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P4_LongestTrips> hdf5 dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/P4_LongestTrips
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-13 23:19 /NYC-Yellow-Taxi-Analysis/output/pig/P4_LongestTrips/_SUCCESS
-rw-r--r-- 1 mdsakib supergroup 594 2026-09-13 23:19 /NYC-Yellow-Taxi-Analysis/output/pig/P4_LongestTrips/part-r-00000
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P4_LongestTrips> hdf5 dfs -cat /NYC-Yellow-Taxi-Analysis/output/pig/P4_LongestTrips/part-r-00000
2016-03-10 07:40:55,2016-03-10 08:39:06,1,44.49,225.0,285.36
2016-03-01 00:00:00,2016-03-01 00:00:00,5,30.43,98.0,113.8
2016-03-10 07:23:27,2016-03-10 08:35:37,6,29.32,52.0,70.01
2016-03-10 07:36:42,2016-03-10 08:29:31,1,29.3,125.5,151.56
2016-03-10 07:32:37,2016-03-10 08:18:50,2,29.22,52.0,63.36
2016-03-10 07:18:41,2016-03-10 08:29:29,1,26.27,75.0,97.61
2016-03-10 07:25:36,2016-03-10 08:13:46,2,25.82,87.5,125.38
2016-03-10 07:25:36,2016-03-10 08:13:43,1,25.21,70.0,89.88
2016-03-10 07:25:28,2016-03-10 08:59:06,1,23.86,52.0,58.34
2016-03-10 07:14:13,2016-03-10 08:50:36,1,22.75,52.0,65.34
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis\pig\P4_LongestTrips>
```

File contents

```
2016-03-10 07:40:55,2016-03-10 08:39:06,1,44.49,225.0,285.36
2016-03-01 00:00:00,2016-03-01 00:00:00,5,30.43,98.0,113.8
2016-03-10 07:23:27,2016-03-10 08:35:37,6,29.32,52.0,70.01
2016-03-10 07:36:42,2016-03-10 08:29:31,1,29.3,125.5,151.56
2016-03-10 07:32:37,2016-03-10 08:18:50,2,29.22,52.0,63.36
2016-03-10 07:18:41,2016-03-10 08:29:29,1,26.27,75.0,97.61
2016-03-10 07:25:36,2016-03-10 08:13:46,2,25.82,87.5,125.38
2016-03-10 07:25:36,2016-03-10 08:13:43,1,25.21,70.0,89.88
```

Pig-5 : Find total revenue and average trip distance by payment type

p5_payment_analysis Script :

```
p5_payment_analysis - Notepad
File Edit Format View Help
taxi_data = LOAD '/NYC-Yellow-Taxi-Analysis/input/taxi_noheader.csv'
USING PigStorage(',')
AS (
  VendorID:int,
  pickup_datetime:chararray,
  dropoff_datetime:chararray,
  passenger_count:int,
  trip_distance:double,
  pickup_longitude:double,
  pickup_latitude:double,
  RatecodeID:int,
  store_and_fwd_flag:chararray,
  dropoff_longitude:double,
  dropoff_latitude:double,
  payment_type:int,
  fare_amount:double,
  extra:double,
  mta_tax:double,
  tip_amount:double,
  tolls_amount:double,
  improvement_surcharge:double,
  total_amount:double
);

valid_data = FILTER taxi_data
BY total_amount >= 0
AND trip_distance >= 0;

grouped_payment =
  GROUP valid_data BY payment_type;

payment_analysis =
  FOREACH grouped_payment GENERATE
    group AS payment_type,
    COUNT(valid_data) AS total_trips,
    SUM(valid_data.total_amount) AS total_revenue,
    AVG(valid_data.trip_distance) AS average_distance;

ordered_result =
  ORDER payment_analysis BY total_revenue DESC;

DUMP ordered_result;

STORE ordered_result
INTO '/NYC-Yellow-Taxi-Analysis/output/pig/P5_PaymentAnalysis'
USING PigStorage(',');
```

Execution :

```
payment_analysis[33,4],grouped_payment[30,4] C: payment_analysis[33,4],grouped_payment[30,4] R: payment_analysis[33,4]
2026-09-14 00:34:24,518 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 0% complete
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0020]
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 16% complete
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0020]
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 33% complete
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Running jobs are [job_1789309900849_0020]
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - 100% complete
2026-09-14 00:34:24,519 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Application state is completed. FinalApplicationStatus=SUCCEEDED. Redirecting to job history server
2026-09-14 00:35:02,796 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 0 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-14 00:35:05,863 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 1 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
```

```
2026-09-14 01:00:34,426 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 8 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-14 01:00:37,464 [main] INFO org.apache.hadoop.ipc.Client - Retrying connect to server: 0.0.0.0/0.0.0.0:10020. Already tried 9 time(s); retry policy is RetryUpToMaximumCountWithFixedSleep(maxRetries=10, sleepTime=1000 MILLISECONDS)
2026-09-14 01:00:39,605 [main] WARN org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MapReduceLauncher - Unable to retrieve job to compute warning aggregation.
2026-09-14 01:00:39,605 [main] INFO org.apache.pig.Main - Pig script completed in 52 minutes, 45 seconds and 379 milliseconds (3163379 ms)
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig P5_PaymentAnalysis> hdfs dfs -ls /NYC-Yellow-Taxi-Analysis/output/pig/P5_PaymentAnalysis
Found 2 items
-rw-r--r-- 1 mdsakib supergroup 0 2026-09-14 00:45 /NYC-Yellow-Taxi-Analysis/output/pig/P5_PaymentAnalysis/ SUCCESS
-rw-r--r-- 1 mdsakib supergroup 102 2026-09-14 00:45 /NYC-Yellow-Taxi-Analysis/output/pig/P5_PaymentAnalysis/part-r-000000
```

Output:

```
PS C:\Users\Md Sakib\Desktop\NYC-Yellow-Taxi-Analysis> pig P5_PaymentAnalysis> hdfs dfs -cat /NYC-Yellow-Taxi-Analysis/output/pig/P5_PaymentAnalysis/part-r-000000
1,4387,72823.30000000211,3.1470207431046164
2,1607,18561.599999999475,2.410342252644679
3,2,30.1,3.85
```

File contents

```
1,4387,72823.30000000211,3.1470207431046164
2,1607,18561.599999999475,2.410342252644679
3,2,30.1,3.85
```

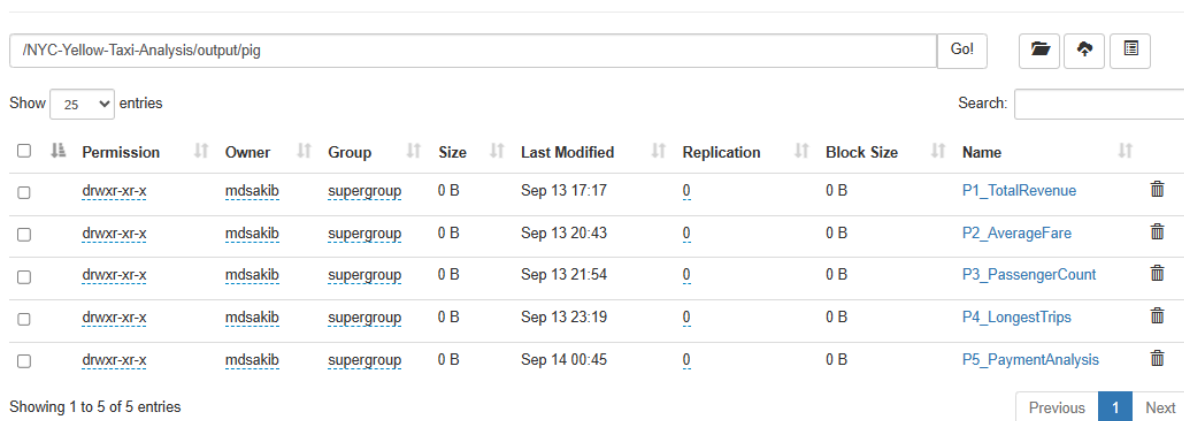
Final Results, Project Verification and Conclusion

The project demonstrates a complete Big Data workflow using a large transaction dataset. The dataset is stored in HDFS, analyzed through five independent MapReduce jobs, and further analyzed through five independent Apache Pig jobs. Separate output folders make each stage easy to verify and present.

Final Localhost Verification

The Hadoop localhost interface can be used to demonstrate the final HDFS structure and the individual output directories. The YARN interface can be used to demonstrate successful application execution.

Hadoop localhost showing /NYC-Yellow-Taxi-Analysis/output/pig with the five Pig task folders



☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	☐
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 13 17:17	0	0 B	P1_TotalRevenue	☐
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 13 20:43	0	0 B	P2_AverageFare	☐
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 13 21:54	0	0 B	P3_PassengerCount	☐
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 13 23:19	0	0 B	P4_LongestTrips	☐
☐	drwxr-xr-x	mdsakib	supergroup	0 B	Sep 14 00:45	0	0 B	P5_PaymentAnalysis	☐

Showing 1 to 5 of 5 entries

Previous 1 Next

Analysis and Discussion

The MapReduce operations demonstrate how Java-based Hadoop processing can be used for counting, grouping, and numerical aggregation. The Pig operations provide higher-level data-analysis capabilities through a concise data-flow language. Together, the operations show how taxi-trip data can be filtered, grouped, sorted, counted, and aggregated to obtain business-relevant information.

The payment-type analysis is particularly useful because it combines several metrics in one result: number of trips, total revenue, and average trip distance. The passenger-count analysis can reveal the most common taxi occupancy pattern, while the longest-trip analysis helps identify unusually long journeys.

Conclusion

This project successfully demonstrates practical big-data analysis using Hadoop MapReduce and Apache Pig on NYC Yellow Taxi trip data. Three MapReduce operations and five meaningful Pig operations were designed to answer different analytical questions. The experiment demonstrates how distributed-data technologies can transform raw taxi records into useful statistical and business information.

References

1. **Kaggle — NYC Yellow Taxi Trip Data — March 2016**
https://www.kaggle.com/datasets/elemento/nyc-yellow-taxi-trip-data?select=yellow_tripdata_2016-03.csv
2. **Java Documentation**
<https://docs.oracle.com/en/java/>
3. **Apache Hadoop Documentation**
<https://hadoop.apache.org/docs/>
4. **Apache Pig Documentation**
<https://pig.apache.org/docs/>
5. **GitHub — NYC_Yellow_Taxi_Trip_Data_Analysis_using_Hadoop Big Data Analytics Project**
https://github.com/sakibskb143/NYC_Yellow_Taxi_Big_Data_Analysis_using_Hadoop